$L(\mathbb{F}_2) \cong L(\mathbb{F}_3)$

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Introduction

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OpenAI release a series of manuscripts claming to sove 300+ problems.

The results can be found on openai's github.

https://github.com/openai/math/tree/main

The results are somewhat astonishing and they span from different aspects of mathematics that are all really deep.

My main goal in this presentation is to introduce one of the problem it solved and its related areas.

Some Thing from OpenAI

An isomorphism of the free group factors

OpenAI

September 23, 2026

Abstract

We solve the free group factor isomorphism problem affirmatively by proving that $L(\mathbb{F}_2)$ and $L(\mathbb{F}_3)$ are isomorphic as tracial von Neumann algebras. The classical free group factor alternative then implies that all interpolated free group factors, including $L(\mathbb{F}_\infty)$, are isomorphic and have fundamental group $\mathbb{R}_{>0}$.

This is the article we are talking about!

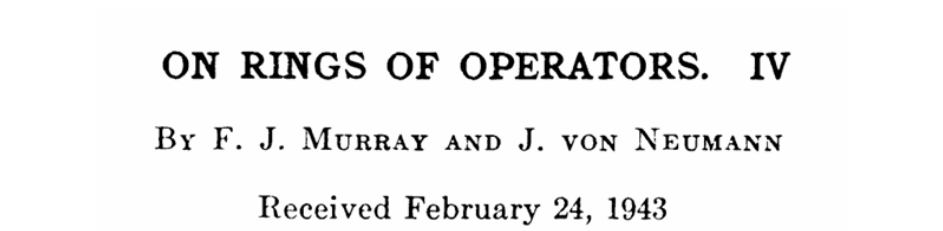
It is listed as “Result 287” out of all 372 results.

This problem is mainly talking about the isomorphism of the free group factors (as the title has said).

Background

To understand the question. We will scan through the history of this problem and introduce the background knowledge (to my best) without introducting the actual tedious proofs.

This problem traces back to John von Neumann and Francis Murray, who sought to provide a rigorous mathematical framework for quantum mechanics. Their efforts ultimately led to the development of von Neumann algebras (or W*-algebras).



ON RINGS OF OPERATORS. IV

By F. J. Murray and J. von Neumann

Received February 24, 1943

This is our starting point...

Background (ii)



Projections

While investigating the bounded linear opeartors, he found that it is important to characterize them through the projections! Since he shows that every operator can be reconstructed through the "projections"

$$A = \int_{\infty}^{\infty} \lambda dE_{\lambda}$$

Thus, if we want to investigate the subspace of the operators, we should ensure that this subspace contains "enough" projections!

Note that we can view this as an analog of the "Spectral decomposition"

$$A = \sum_{i=1}^{n} \lambda_i P_i$$

The idea of investigating operators become investigating the "projections".

von Neumann algebra

The common definition of von Neumann algebra is as follows:

*The weakly closed *-algebras of bounded operators (on a Hilbert space) containing the identity.*

Interestingly, von Neuamnn has discovered that this definition is equivalent to

The bicommutant of M is equal to itself. (von Neumann 1930)

Of course, similar to C^* algebra, there is a more abstract/ algebraic version of definition

C^* algebras that has predual (Sakai 1971)

Note: von Neumann algebra is of course a special type of C^* algebra which is closed in norm topology.

Factors

A von Neumann algebra N whose center consists only of multiples of the identity operator is called a factor.

For example, $M_n(\mathbb{C})$ is a factor while $M_2(\mathbb{C}) \oplus M_2(\mathbb{C})$ is not a factor. The resason is that $\lambda I_4 \neq aI_2 \oplus bI_2 \in Z(M_2(\mathbb{C}) \oplus M_2(\mathbb{C}))$.

von Neumann and Murray (1936) showed that there are only 3 types of factors.

- type I: There is a minimal projection $E \neq 0$.
- type II: There are no minimal projection but there is a a nonzero finite projection.
- type III: There are no finite projection.

von Neumann (1940) later give the complete factorization of any von Neumann algebra. He showed that every von Neumann algebra of a seperable Hilbert space is isomorphic to the direct sum of factors.

Projections II

We have mentioned that projection serve as an important role in operator algebra. Thus, we would like to understand projection (or the subspace behind itself) first.

One way that we can classify projection is through their image's dimensionality, e.g. in $M_3(\mathbb{C})$, we can say

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \cong \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

since they project to the subspaces with same dimension (which is 2). Thus, we can use "Tr" to classify them.

However, if the subspace is infinite, this method cannot classify them.

Thus, we should use "normalized" trace "$\mathrm{tr} := \frac{1}{n}\ \mathrm{Tr}$" instead.

In this case, we can not measure the finite dimensional (all becomes 0).

Projections II (ii)

The generalization of this idea is through MvN equivalence which states that

$$p \sim q \in M \Leftrightarrow p = uu^*, q = u^*u \text{ for some partial isometry } u \in M$$

This equivalent can induces a natural partial order of projections.

For *finite projection* p, we define as there is no projection $q < p$ with $p \sim q$.

For *minimal projection* p, we define as there is no smaller projection, i.e. $0 < q < p$

More importantly, we notice that this concept doesn't use "tr" to define the dimensionality.

Type I

Lets look at some examples of Type I factors.

The most simple example is the matrix algebra $M_n(\mathbb{C}) \cong B(\mathbb{C}^n)$. It is easy to see that there is the *minimal projection* of such algebra, for example,

$$Pe_i = \begin{cases} e_1 & \text{if } i = 1 \\ 0 & \text{if } i > 1 \end{cases}$$

where e_i is the basis of $\mathbb{C}^n$.

Now, consider the seperable Hilbert space H, we have $B(H)$ also a type I von Neumann algebra.

Type II_1

There are two types of Type II factors. If the identity operator in a type II factor is finite, the factor is said to be of type II_1; otherwise, it is said to be of type II_∞.

Lets again look at some example of Type II_1. Consider the following sequence

$$M_{2^i} = M_{i-1} \otimes M_2; M_2 = M_2(\mathbb{C})$$

Then, define

$$R = \overline{\bigcup_i M_{2^i}} \subset B(H)$$

R is a Type II_1 factor, as we can define $\tau_n(X) = \frac{1}{2^n} \operatorname{Tr}(X)$. Then, one can show that $\tau_n(m_n \oplus m_n) = \tau_{n-1}(m_n)\tau_1(I_2)$ and thus we can define a normalized "trace" on R.

Intuitively, we know that there is no *minimal projection*, as we always can find the "half" space. R is called hyperfinite type II factor.

Free group factor

Another example of Type II_1 von Neumann algebra is the free group factor.

To introduce this, we first introduce the group algebra.

Consider a discrete group G and a compactly supported continuous function $a : G \to \mathbb{C}$ with convolution $(a, b) \mapsto a * b$. Then, the set of such functions can be identified with the group algebra $\mathbb{C}(G)$ as the formal linear combinations of elements in G with complex coefficients, i.e. $a = \sum_{g \in G} a(g)g \in \mathbb{C}(G)$. The operation is given by $a * b := \sum_{g \in G} (a * b)(g)g \in \mathbb{C}(G)$ and the identity is given by $\delta_e = 1 \cdot e$.

We can also give $\mathbb{C}(G)$ some "structure". Lets define the inner product on it.

$$\langle g, h \rangle := \begin{cases} 1 \text{ if } g{=}h \\ 0 \text{ if } g{\neq}h \end{cases}$$

We can extend this inner product on G to $\mathbb{C}(G)$. Then, we have the induced $(\mathbb{C}(G), \|\cdot\|_2)$ as our normed space (and a Hilbert space).

Free group factor (ii)

Note that if G is infinite, we have to make sure that $\sum_{g\in G}|a(g)|^2 < \infty$ and taking the completion! Such space is denoted by $l^2(G)$.

Now, consider $\lambda : G \to \mathcal{U}(l^2(G))$ by

$$\lambda(h)\sum_{g\in G} a(g)g \mapsto \sum_{g\in G} a(g)hg$$

Since $\langle hg, k\rangle = \langle g, h^{-1}k\rangle$, we have $\lambda(h)^* = \lambda(h^{-1})$. Also, because λ is a group homomorphism, we have $\lambda(g)\lambda(g)^* = I = \lambda(g)^*\lambda(g)$. We have that $\lambda(g)$ is indeed a unitary of $l^2(G)$.

We can easily extend the domain of λ to $\mathbb{C}(G)$ by

$$\lambda\left(\sum_{g\in G} a(g)g\right) = \sum_{g\in G} a(g)\lambda(g)$$

Then, the result is that $\lambda : \mathbb{C}(G) \to B(l^2(\mathbb{G}))$ is an algebra homomorphism.

Free group factor (iii)

Finally, we define the group von Neumann algebra $L(G)$ by

$$L(G) := \overline{\lambda(\mathbb{C}(G))}^{\mathrm{WOT}} \subset B(l^2(G))$$

This is indeed a von Neumann algebra.

We can find the trace defined by $\tau\left(\sum_{g\in G} a(g)\lambda(g)\right) := \sum_{g\in G} a(g)\langle\lambda(g)e, e\rangle$

- $\tau(xy) = \sum_{g,h\in G} a(g)b(h)\langle gh, e\rangle = \sum_{g,h\in G} b(h)a(g)\langle hg, e\rangle = \tau(yx)$
- $\tau(\lambda(e)) = \langle e, e\rangle = 1$

Then, if the group von Neumann algebra is a factor with no minimal projection. We have $L(G)$ is a type II_1 factor.

This is indeed true for the nontrivial group having ICC (Infinite Conjugacy Class) property and in turn a type II_1 factor.

Now, consider the free group F_n. It has the ICC property (for $n \geq 2$) and thus $L(F_n)$ is a type II_1 factor.

Free group factor (iv)

Murray and von Neumann showed that $L(F_n)$ is not isomorphic to hyperfinite factor. Thus, it is some new structures. However, nothing was known about the structure of these free group factors.

While F_n is isomorphic to F_m only when $n = m$, can we expect $L(F_n) \cong L(F_m)$? This problem is the free group factor isomorphic problem.

> Free group factor isomorphism: Let $m, n \geq 2$ (possibly equal to ∞) and $n \neq m$. Are von Neumann algebrs $L(F_n) \cong L(F_m)$?

To address this issue, Voiculescu invented the "free" probability to understand them better.

Free probability

While investigating the free group factor, Voiculescu see $L(F_n)$ as some kind of "probability space", as we have normalized trace τ (expectation) and the underlying space ($L(F_n)$). The last thing he does is to notice the "independence" of different elements, which he call it "free independence" (1985).

He also notice that random matrices also obey this kind of "free independence" as $n \to \infty$ (1991), which he calls it "asymptotic freeness". This make the problem more feasible as we can actually deal with "free" objects from finite random settings and taking the limit.

His students Dykema later (1992) showed that: We must have either

- $L(F_n) \cong L(F_m)$ for all $1 < n, m < \infty$ and the fundamental group of $L(F_n)$ is $\mathbb{R}_+$ for all $1 < n < \infty$
- $L(F_n) \ncong L(F_m)$ for all $1 < n, m < \infty$ and the fundamental group of $L(F_n)$ is $\{1\}$ for all $1 < n < \infty$

Free probability (ii)

This make the isomorphism problem becomes of finding whether

$$L(F_2) \cong L(F_3)?$$

This was claimed to be solved by OpenAI by proving this was right!

Maybe closing this problem?

Theorem 1.2. *There exists a unital normal trace-preserving $*$-isomorphism*

$$\Phi : L(\mathbb{F}_2) \longrightarrow L(\mathbb{F}_3).$$

This gives an affirmative solution to the free group factor isomorphism problem. Together with the classical dichotomy, this proves that all interpolated free group factors, including the infinite-rank factor, are isomorphic. Their fundamental groups are therefore $\mathbb{R}_{>0}$. We state and prove these